

Demonstrating Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality

Shan-Yuan Teng
University of Chicago, USA

Pengyu Li
University of Chicago, USA

Romain Nith
University of Chicago, USA

Joshua Fonseca
University of Chicago, USA

Pedro Lopes
University of Chicago, USA



Figure 1: We engineered a *foldable* haptic device worn on the user’s fingernail that renders touch in mixed reality (MR) without preventing users from *also* touching real objects. Here, a user follows MR instructions to repair their bicycle. (a) When the user touches the virtual tire, a cover slides down from our nail-device and pushes against their fingerpad to create the *contact & pressure* with this virtual object. To further increase realism, the cover also includes a linear resonant actuator that renders virtual textures using vibrations. This allows, for instance, to feel the roughness of the virtual mountain tire. Then, (b) when the user turns to interact with real objects, the cover folds back, leaving their fingerpads free to feel the texture of their real tire or operate physical tools.

ABSTRACT

We propose a nail-mounted *foldable* haptic device that provides tactile feedback to mixed reality (MR) environments by pressing against the user’s fingerpad when a user touches a virtual object. What is novel in our device is that it quickly tucks away when the user interacts with real-world objects. Its design allows it to fold back on top of the user’s nail when not in use, keeping the user’s fingerpad free to, for instance, manipulate handheld tools and other objects while in MR. To achieve this, we engineered a wireless and self-contained haptic device, which measures 24×24×41 mm and weighs 9.5 g. Furthermore, our foldable end-effector also features a linear resonant actuator, allowing it to render not only touch contacts (i.e., pressure) but also textures (i.e., vibrations). We demonstrate how our device renders contacts with MR surfaces, buttons, low- and high-frequency textures.

ACM Reference Format:

Shan-Yuan Teng, Pengyu Li, Romain Nith, Joshua Fonseca, and Pedro Lopes. 2021. Demonstrating Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI ’21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451540>

1 INTRODUCTION

Mixed Reality (MR) allows overlaying digital content with our real-world surroundings, creating powerful new tools. Many argue that the next challenge of mixed reality is the addition of haptics. Over the last decades, an impressive number of haptic devices have allowed users to feel the forces (e.g., exoskeleton gloves [2, 12]) and contacts from interacting with virtual objects (e.g., vibration on the fingerpads [7]). However, researchers argue that haptics for MR are inherently different from haptics for virtual reality (VR), as they must leave the user’s hands free so that the user can also interact with real objects [8, 11].

This is imperative as many powerful uses of MR involve holding tools such as for repairs [4] or tangible interactions [5]. Therefore, haptics for MR must not impair the user’s ability to feel the world through their fingerpads. Recently, researchers proposed unencumbered force-feedback in MR by using electrical muscle stimulation instead of the traditional exoskeletons [8]. However,

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI ’21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451540>

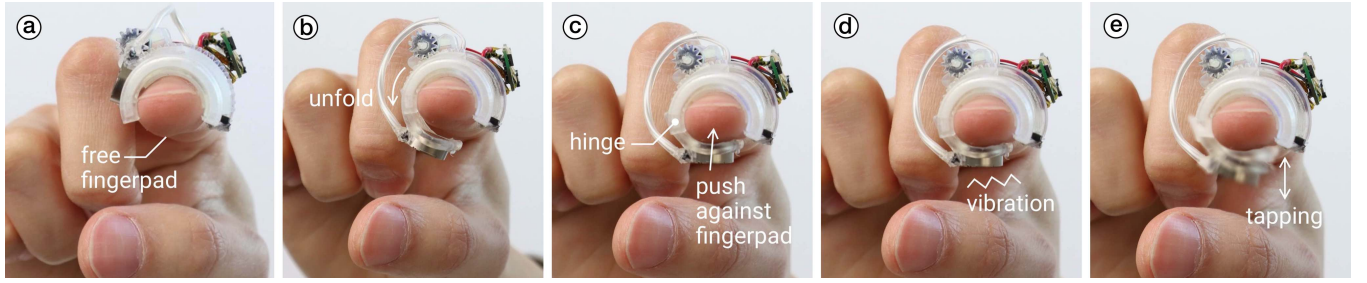


Figure 2: (a) When not in use our device keeps the fingerpad free for feeling the haptics of real-world objects. When activated, it unfolds (b) via a rack and pinion, and (c) a hinge redirects the force against the user’s fingerpad, which allows it to create three types of haptic effects: contact (pressure of the cover), (d) high-frequency textures (using an LRA embedded in its cover), and (e) low-frequency textures (by rocking the cover back and forth).

this only provides the sense of pushing against an object and does not stimulate the sense of touching an object.

As of now, there is no device that provides users with a sense of contact at the fingerpad in MR without covering up their fingerpads. Existing approaches involve applying actuators on the fingerpads, such as thin electrodes [6, 11] or soft actuators [3]. While some of these are also driven by the goal of minimal interference with the user’s tactile sensation, adding these thin patches decreases one’s ability to perform discriminate textured surfaces because these patches impair the tactile acuity of our fingerpads [9].

When it comes to rendering touch in MR without covering up the fingerpads, the most promising solution remains placing a vibration motor on the user’s fingernail [1]. While this leaves the fingerpad free, it has two main disadvantages: it does not create pressure, and its feedback is unrealistic, as it occurs on the fingernail rather than on the fingerpad.

We tackle this challenge by engineering a foldable haptic device that provides virtual objects with haptic feedback by pressing against the user’s fingerpad, yet, quickly tucking away when the user grasps real objects. Our device, depicted in Figure 1, works by unfolding a cover that wraps around and presses against the user’s fingerpad. The key to its compact form factor is that the unfolding cover can be retracted and stored on top of the fingernail via a motor-driven rail. Furthermore, besides rendering the sense of touch, it also renders textures by means of a linear resonant actuator (LRA) embedded in its cover, as depicted in Figure 2

Besides being a one-of-a-kind device for MR haptics, it is also completely *untethered* and *self-contained*, a feature not seen in any existing haptic device of this kind. In its small footprint (24×24×41 mm and 9.5 g), it includes actuators, electronics, battery, and wireless communication through Bluetooth. We demonstrate how our device renders haptic sensations such as taps, button presses, and low- or high- frequency textures.

2 SYSTEM WALK-THROUGH

We demonstrate the key qualities of our device with the example of a MR application for bike repairs. Here, the user wears our nail-mounted device as well as a HoloLens 2 (Microsoft), which displays the MR content and tracks their hands via built-in depth cameras.

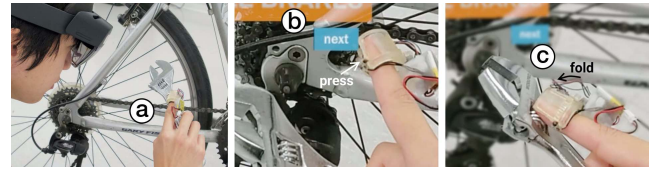


Figure 3: (a) The user holds a wrench while wearing our device. (b) When the user reaches out to touch the virtual button, our device unfolds and renders the contact with the MR button; then (c) as soon as the user’s finger leaves the MR interface, our device folds back, allowing the user to feel and manipulate the wrench with their bare fingerpad.

2.1 Keeping the User’s Fingers free to Manipulate Tools

The user cycles between interacting with the MR repair guide and working with physical tools (Figure 3a). The user taps the mid-air “next” button to reveal the instructions for removing the wheel hub nuts (Figure 3b). Immediately, our device unfolds and pushes its cover against the user’s fingerpad, rendering the pressure of the contact with the MR button. Our device takes 92 ms to unfold. Then, as soon as the user’s finger leaves the MR interface, the cover retracts backward, leaving the user’s fingerpad free to adjust the wrench’s jaw while still even wearing our device, as depicted in Figure 3c. Note this is a dexterous task, and would not be possible with existing haptic devices, as they would interfere with the user’s tactile acuity and disrupt the task.

Now that the wheel has been removed, the MR guide displays two different virtual tire profiles (mountain tires vs. racing tires). The user compares their tire with the virtual tires. Our device not only keeps the user’s fingerpad free to feel their actual tire’s texture (Figure 4a) but also unfolds to creates the textures of different virtual tires.

In fact, our device renders two different textures for two different virtual tires. First, as depicted in Figure 4b, the user strokes their finger across the virtual mountain tire; our device unfolds its cover around the fingerpad and quickly rocks the cover back and forth to render the low-frequency texture of the tire bumps. Then, as depicted in Figure 4c, the user feels the profile of the virtual racing

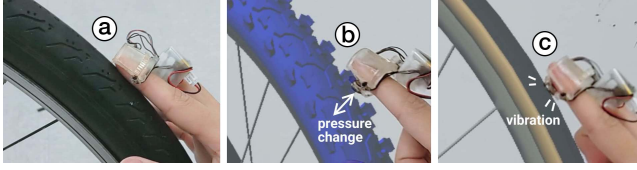


Figure 4: (a) The user feels the texture of their real tire. (b) Our device moves its cover back and forth to render the bumps of a rough, virtual mountain tire. (c) Our device vibrates its LRA to render the fine texture of a virtual racing tire.



Figure 5: (a) User grips the bike handle while wearing our device. (b) As they type the address, our device renders the clicking of the keys. (c) When they drag the slider, our device renders haptic detents that depict the magnification levels.

tire. Here, our device unfolds and pushes the cover against the user’s fingerpad. Then, as the user strokes across the tire, the embedded vibration actuator (LRA) renders the fine, high-frequency texture typical of a racing tire.

2.2 Providing Haptics to General MR Widgets

After replacing the tires, the user bikes to their friend’s house using the MR interface for driving directions. Figure 5a depicts the user grasping the bike’s handle, while still wearing our device. When the user types in the address on a virtual keyboard, our device creates contact feedback for each keystroke by unfolding and pressing its cover against the user’s fingerpad (Figure 5b). When the direction is being shown in a virtual map, the user zooms using the map’s slider. Here, our device provides not only contact but also haptic detents using vibration motor to depict the different magnification levels (Figure 5c).

3 IMPLEMENTATION

Figure 6 depicts our self-contained prototype¹, which was 3D printed using a Form Labs 3 with clear resin to minimize visual interference with the real world. Its complete footprint is 24×24×41 mm and weighs 9.5 g, including its battery. It attaches to the user’s fingernail using double-sided tape.

3.1 Folding Mechanism

At the core of our contribution is our folding mechanism, depicted in Figure 7. Its key design feature is a hemispheric rail from which

our “cover” unfolds. Once extended, the cover presses against the user’s fingerpad.

The cover is comprised of two segments connected via a thin plastic sheet. To fold or unfold, we actuate the cover using a rack and pinion drive. Figure 7a depicts the initial configuration, in which the cover’s front segment stays in the case. Figure 7b depicts the cover’s front segment is pushed as it is driven by the pinion to the point where the front segment is fully extended and hits a hinge stopper at the end of the rail. Figure 7c depicts the last stage in which a wedge in the cover pushes and causes the front segment to pivot around the hinge and land flush against the fingerpad. The shape of our casing, cover and its hemispherical rail are all conical in order to ergonomically follow the finger’s shape, allowing the cover to fully contact with the fingerpad.

3.2 Actuation: Pressure and Vibration

Our device unfolds using a DC motor (26:1 Sub-Micro Planetary Gearmotor 0.1 kg-cm, Pololu) mounted on our 3D printed rail drive (rack with 26 teeth and pinion with 12 teeth). To increase the expressivity of our device, we embedded a linear resonant actuator (LRA C10-100, Precision Micro Drives) in the cover that touches the user’s finger. Our LRA renders high-frequency textures, allowing our device to render both contact pressure and a wider range of textures. We drive our LRA using a MOSFET between 150-190 Hz; its resonant frequency is at 170Hz. Lastly, a force sensor (FSR 400, Interlink Electronics) and a photo interrupter (SG-105F, Kodenshi) close the control loop by acting as limit switches. The force sensor also doubles as a feedback signal for fine-tuning the amount of pressure applied on the fingerpad. Thus, our device not only creates pressure but also constantly measures the applied force. Finally, we used the photo interrupter to sense whenever the cover is fully retracted, which serves as the signal to stop actuating our DC motor.

3.3 Electronics: Printed Circuit Board and Schematics

Our 16.8x10.3 mm PCB houses at its core a microcontroller with Bluetooth Low Energy (nRF52811, Nordic Semiconductor). To decrease its footprint, we used a ceramic chip antenna (W3008C, Pulse Larsen), instead of the traditional zig-zag PCB antennas. We power our device using a 40 mAh LiPo battery. We measured a current of 200 mA when it unfolds, which takes 184 ms per interaction. As such, our device can be used for 12 min of continuous tactile feedback. It is worth noting that in typical interactions with MR interfaces one just expects to feel a few hundred milliseconds of contact (e.g., tapping a button), thus our device’s battery tends to last for many hours of on-demand use.

3.4 Tracking and Display

To display the graphics to the user, we used a HoloLens 2. Built-in depth cameras on the headset are used to track their hands. We unfold our device to tap the user’s fingerpad only when the finger collides with a virtual object. Then, whenever the user’s finger leaves the virtual object’s collider (the “touch” area programmed in our Unity3D demos), we immediately retract the cover back onto the fingernail to leave the user’s fingerpad free. We implemented

¹More details can be found in our paper [10]. We also provide all the source code, firmware and schematics at <http://lab.plopes.org/#touchfold>

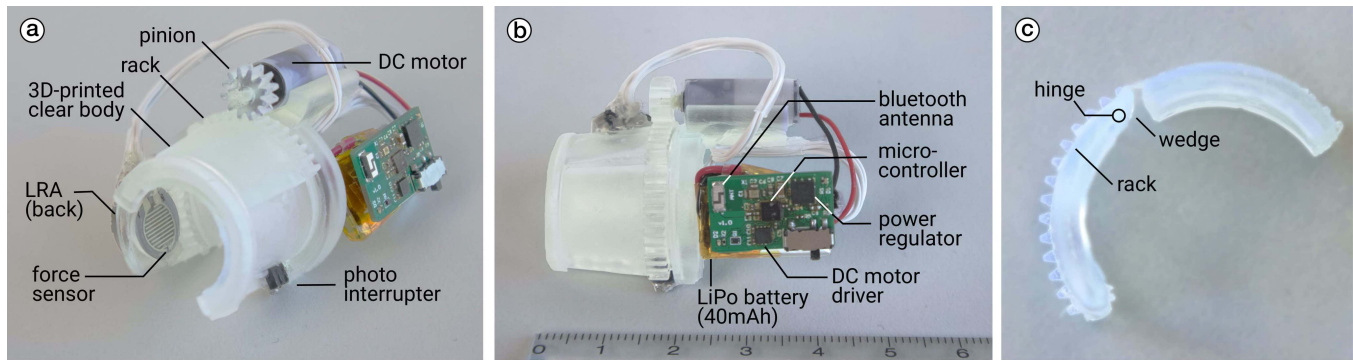


Figure 6: (a) Our self-contained haptic device; (b) detail of its cover; (c) top view of our device with ruler unit in cm.



Figure 7: The key to our mechanism's unfolding is that (a) its cover is comprised of two segments; (b) our hinge is based on two detents that create a stopper and (c) force the cover to pivot and be pushed by the wedge against the fingerpad.

this strategy as it is ideal for MR, since it prioritizes real-world interactions over virtual interactions. To trigger our device when the user touches an MR object, we expand the finger's collision box in Unity3D to the radius of our device, which further compensates for its aforementioned latency.

4 LIMITATIONS & CONCLUSION

We proposed the first foldable, nail-mounted haptic device that provides tactile feedback to mixed reality (MR) experiences while *quickly tucking away* when the user interacts with real-world objects. To achieve this unencumbered haptic feedback, we engineered a wireless haptic device, which measures 24×24×41mm and weighs 9.5g. Furthermore, our foldable end-effector also features a linear resonant actuator allowing it to render not only touch contacts (i.e., pressure) but also textures (i.e., vibrations). Our device is not without its limitations: (1) while we optimized its footprint, it still covers up the user's fingernail, and obstructs the side of the fingertip at certain angles; (2) while we fabricated it to fit the average adult finger, it might need adjusting for different finger sizes; lastly, (3) as with any mechanical haptic device, it has its inherent latency (92ms), which we currently compensate by enlarging the collision detector volumes in our MR applications. We believe that our device might inspire researchers to explore new types of foldable haptic designs

that optimize their form-factor and are therefore more suited for MR.

REFERENCES

- [1] Hideyuki Ando, Eisuke Kusachi, and Junji Watanabe. 2007. Nail-mounted Tactile Display for Boundary/Texture Augmentation. In *Proceedings of the International Conference on Advances in Computer Entertainment Technology (ACE '07)*, 292–293. <https://doi.org/10.1145/1255047.1255131>
- [2] Inrak Choi and Sean Follmer. 2016. Wolverine: A Wearable Haptic Interface for Grasping in VR. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16 Adjunct)*, 117–119. <https://doi.org/10.1145/2984751.2985725>
- [3] Teng Han, Fraser Anderson, Pourang Irani, and Tovi Grossman. 2018. HydroRing: Supporting Mixed Reality Haptics Using Liquid Flow. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*, 913–925. <https://doi.org/10.1145/3242587.3242667>
- [4] Steven J. Henderson and Steven Feiner. 2009. Evaluating the benefits of augmented reality for task localization in maintenance of an armored personnel carrier turret. In *2009 8th IEEE International Symposium on Mixed and Augmented Reality*, 135–144. <https://doi.org/10.1109/ISMAR.2009.5336486>
- [5] Anuruddha Hettiarachchi and Daniel Wigdor. 2016. Annexing Reality: Enabling Opportunistic Use of Everyday Objects as Tangible Proxies in Augmented Reality. 1957–1967. <https://doi.org/10.1145/2858036.2858134>
- [6] Hiroyuki Kajimoto. 2012. Electrotactile Display with Real-Time Impedance Feedback Using Pulse Width Modulation. *IEEE Transactions on Haptics* 5, 2: 184–188. <https://doi.org/10.1109/TOH.2011.39>
- [7] Hwan Kim, Minhwan Kim, and Woohun Lee. 2016. HapThimble: A Wearable Haptic Device Towards Usable Virtual Touch Screen. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*, 3694–3705. <https://doi.org/10.1145/2858036.2858196>
- [8] Pedro Lopes, Sijing You, Alexandra Ion, and Patrick Baudisch. 2018. Adding Force Feedback to Mixed Reality Experiences and Games Using Electrical Muscle Stimulation. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*, 446:1–446:13. <https://doi.org/10.1145/3173574.3174020>
- [9] Aditya Shekhar Nittala, Klaus Krutwig, Jaeyeon Lee, Roland Bennewitz, Edward Arzt, and Jürgen Steimle. 2019. Like A Second Skin: Understanding How Epidermal Devices Affect Human Tactile Perception. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*, 380:1–380:16. <https://doi.org/10.1145/3290605.3300610>
- [10] Shan-Yuan Teng, Pengyu Li, Romain Nith, Jersey Fonseca, and Pedro Lopes. 2021. Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. <https://doi.org/10.1145/3411764.3445099>
- [11] Anusha Withana, Daniel Groeger, and Jürgen Steimle. 2018. Tacttoo: A Thin and Feel-Through Tattoo for On-Skin Tactile Output. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*, 365–378. <https://doi.org/10.1145/3242587.3242645>
- [12] CyberGrasp. *CyberGlove Systems LLC*. Retrieved March 8, 2020 from <http://www.cyberglovesystems.com/cyberggrasp>